



Liquid-liquid phase separation causes high turbidity and pressure during low pH elution process in Protein A chromatography



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ABSTRACT

Turbid elution pools and high column back pressure are common during elution of monoclonal antibodies (mAbs) by acidic pH in Protein A chromatography. This phenomenon has been historically attributed to acid-induced precipitation of incorrectly folded or pH-sensitive mAbs and host cell proteins (HCPs). In this work, we propose a new mechanism that may account for some observations of elution turbidity in Protein A chromatography. We report several examples of turbidity and high column back pressure occurring transiently under a short course of neutral conditions during Protein A elution. A systematic study of three mAbs displaying this behavior revealed phase separation characterized by liquid drops under certain conditions including neutral pH, low ionic strength, and high protein concentration. These liquid droplets caused solution turbidity and exhibited extremely high viscosity, resulting in high column back pressure. We found out that the droplets were formed through liquid-liquid phase separation (LLPS) as a result of protein self-association. We also found multiple factors, including pH, temperature, ionic strength, and protein concentration can affect LLPS behaviors. Careful selection of process parameters during protein A elution, including temperature, flow rate, buffer, and salt can inhibit formation of a dense liquid phase, reducing both turbidity (by 90%) and column back pressure (below 20 pounds per square inch). These findings provide both mechanistic insight and practical mitigation strategies for Protein A chromatography induced LLPS.

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1. Introduction

Owing to their high specificity and strong antigen binding, monoclonal antibodies (mAbs) have emerged as powerful human therapeutics for oncology, inflammation, infectious diseases, autoimmune diseases, and metabolic disorders [1]. Biopharmaceutical companies have established purification platforms to standardize manufacturing processes and expedite process development for mAbs having differing physiochemical properties [2–5]. Although specific details of these platform processes can differ, they typically rely on Protein A affinity chromatography as the capture step [4,5].

Protein A affinity chromatography is based on the strong and specific binding of the Fc region (fragment crystallizable region) of the mAb and a Protein A ligand immobilized on a support matrix. Use of Protein A for mAb capture is proven to be highly selective and can achieve over 99% product purity when starting from a crude cell culture supernatant [4]. The strong affinity of this interaction also enables the usage of relatively strong wash buffers (such as high concentration of sodium chloride (NaCl), arginine hydrochloride (Arg-HCl), urea, isopropanol, etc.) to remove host cell proteins, DNA and other process related impurities [6,7]. Following the wash step, a re-equilibration buffer at neutral pH (for instance, Tris-HCl or sodium phosphate at pH 7.4) is used to remove residual components present in the wash buffer from the Protein A column. This re-equilibration buffer also helps protein recovery from potential stress of the strong wash buffer. In order to decrease the conductivity of the elution product to better integrate with the subsequent unit operations, the re-equilibration buffer usually has minimal or

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no salt [4], such as 50 mM Tris-HCl, pH 7.4 or 50 mM sodium phosphate, pH 7.0. The mAb products bound on the Protein A ligand are then eluted with an acidic buffer such as sodium acetate, glycine or sodium citrate at pH 3.0–4.0 [2,4,8].

Acidic pH conditions required for Protein A chromatography elution generally induces stress on the protein and may lead to aggregation and precipitation of unstable mAbs [8–10]. Inclusion of urea or arginine in the low pH elution buffer or operating Protein A purification at low temperature are effective solutions for preventing the precipitation and aggregation of some mAb products [2,10,11]. Besides the low pH stress, adsorption to or desorption from the Protein A ligand has also been shown to contribute to protein aggregation. Mazzer et al. suggested that Protein A chromatography increases monoclonal antibody aggregation rate during the subsequent low pH viral inactivation step [8]. Gagnon et al. recently showed that Protein A purification reduces the hydrodynamic diameter of the eluted mAb, leading to greater susceptibility to aggregation [12]. In other cases, precipitation during low pH elution in Protein A chromatography was reported to be associated with host cell protein contaminants, rather than the mAb product itself [13]. Re-purification of the Protein A purified product again by the Protein A chromatography can be used to distinguish between precipitation of host cell protein or mAb product [13]. These low pH-induced precipitates of HCPs are usually irreversible, and the elution pool remains turbid prior to being filtered. In process-scale chromatography skids, there are usually several filters (such as depth filter and aseptic filtration for the elution product) in-line with the Protein A column. A turbid elution can clog the Protein A column itself or these filters and cause high back pressure [5], complicating manufacturing operations.

Under certain conditions, such as high protein concentration and low ionic strength, liquid droplets can form in a protein solution and can scatter light resulting in a turbid appearance [14,15]. These liquid droplets contain high concentration of protein and, therefore, are denser than the elution pool solution. Under the influence of gravity, these droplets separate from the rest of the solution, and two liquid phases eventually form with the high protein concentration in the bottom layer [15]. The protein concentration of each layer is dependent on temperature and buffer conditions. This is driven by a process termed liquid–liquid phase separation (LLPS) [15]. In recent years, many mAbs or mAb derivatives have been reported to show LLPS behaviors [14,16–21]. These published studies suggest that the LLPS behaviors of these proteins are reversible upon changing the temperature and buffer conditions [17–19].

Protein A stationary phases used in biopharmaceutical manufacturing now routinely have high binding capacities, often exceeding 30 mg mAb/mL resin [22]. Accordingly, the instantaneous (at local) product concentration can readily exceed 30 mg/mL in the column effluent during the elution process, although the concentration of the final elution product pool is usually lower (typically 10–20 mg/mL). As described above, low ionic strength and low salt concentration solution is often used for elution of mAb from the Protein A; conditions which tend to promote LLPS formation for some mAbs.

In this work, we conducted a systematic study on LLPS induced turbidity and high back pressure during issue in Protein A chromatography. Briefly, we (1) used microfluidic imaging (MFI) to show the presence of liquid droplets and prove the formation of LLPS in the Protein A chromatography; (2) used viscometer and dynamic light scattering instruments to study viscosity and self-association behaviors; (3) constructed liquid–liquid coexistence curve (phase diagram) using T_{cloud} that obtained by UV₆₀₀ transmittance; (4) evaluated the effects of elution conditions and operational parameters in Protein A chromatography.

Table 1

Summary for the screened mAbs and their results from Protein A chromatographic experiments.

mAb	Isotype	pI	Self-association ^a	Elution Turbidity ^b
A	IgG 1	7.9	Yes	Yes
B	IgG1	8.3	Yes	Yes
C	IgG4	6.8	Yes	Yes
D	IgG4	7.6	Yes	Yes
E	IgG2	8.2	Yes	Yes
F	IgG2	8.1	No	No
G	IgG1	8.3	Yes	No
H	IgG1	8.7	Yes	No
I	IgG2	8.7	Yes	Yes
J	IgG1	8.6	No	No
K	IgG1	9.5	Yes	No
L	IgG4	6.3	Yes	No

^a Self-association propensity was evaluated by DLS. “yes”, negative K_D ; “no”, close to zero or positive K_D .

^b The turbidity of the elution was monitored by UV₆₀₀ in AKTA Explorer. “yes”, UV₆₀₀ peak height >50 mAU; “no”, UV₆₀₀ peak height <50 mAU.

2. Materials and methods

2.1. Chemicals, monoclonal antibodies, and Protein A resins

The chemicals used in this study were obtained from J.T. Baker (Phillipsburg, NJ, USA). Buffers and cleaning solutions were prepared in-house. The 12 mAbs evaluated were produced by MedImmune, LLC (Gaithersburg, MD, USA). They are humanized monoclonal antibodies (IgG1, 2, or 4) comprising two identical heavy chains and two identical light chains, with an overall molecular weight of around 150 kDa. The theoretical isoelectric points for these mAbs are summarized in Table 1. Unless noted otherwise, the proteins used in this study were pre-purified with greater than 95% monomer purity and host cell protein levels below 100 ng/mg. MabSelect™ SuRe™ (MSS) Protein A and MabSelect Xtra™ resins were from GE Healthcare (Piscataway, NJ, USA); POROS® MabCapture™ A resins were from Applied Biosystems (Grand Island, NY, USA); Toyopearl AF-rProtein A resins were from Tosoh Bioscience (King of Prussia, PA, USA).

2.2. Chromatography instrument and operations

The chromatographic experiments were carried out on an ÄKTA Explorer 100 controlled by Unicorn software version 6.4 (GE Healthcare, Piscataway, NY, USA). The resins was packed into 1.15 cm inner diameter (ID) Vantage L laboratory columns (Millipore, Billerica, MA, USA) to a bed height of 19 ± 3 cm. Unless mentioned otherwise, purification by Protein A was performed at room temperature under the following conditions: 3 column volume (CV) 50 mM Tris-HCl, pH 7.4 for equilibration before sample loading; 30 mg protein per mL resins as load challenge; 3 CV re-equilibration buffer A for post-loading equilibration; 5 CV 25 mM sodium acetate (NaAc) pH 3.6 for elution; 100 mM acetic acid for column regeneration; 3 CV 0.1 M NaOH for sanitation; and 3 CV 20% ethanol for column storage, where all steps were operated at a flowrate of 300 cm per hour. Other column temperatures were achieved using a temperature-controlled water jacket around the column. The purification process was monitored at UV₂₈₀ and UV₆₀₀ using the in-line ÄKTA sensor. Protein A elution was collected in 0.25 CV per fraction based on a UV₂₈₀ collection criterion of 50 mAU. pH and conductivity were monitored by in-line probes and off-line (for the collected fractions) using a calibrated Mettler-Toledo pH and conductivity meters (Columbus, OH, USA). Sample protein concentration was measured using a NanoDrop™ 2000 (Thermo Fisher Scientific, Wilmington, DE, USA). The column back pressure was found to correlate well with the UV₆₀₀ signal, i.e. the lower UV₆₀₀

peak, the lower the column back pressure. To simplify discussion, the impacts of different factors were described mainly by UV₆₀₀ peak.

2.3. Micro-flow imaging (MFI)

MFI System model DPA 4200 ProteinSimple (San Jose, CA, USA) was used to capture the images of the three turbid fractions for mAb A, B and C. The system was pre-rinsed using the filtered buffer (50 mM Tris-HCl, pH 7.4) before running each sample. For each MFI measurement, 1 mL of degassed sample was loaded into the MFI system using a barrier pipette tip after gently swirling the solution. Images were collected at the flowrate of 100 μ L per minute over 1 min and processed using MFI View analysis suite software version 1.0 (ProteinSimple).

2.4. Liquid-liquid coexistence curve

Opalescence and turbidity of the solution was measured as percent transmittance using an 8453 UV system from Agilent Technologies (Santa Clara, CA, USA). The 8453 UV system was equipped with a temperature control module connected to a water bath. The measurement was carried out at 600 nm (where proteins have no absorbance) in a quartz cuvette with a path length of 1 cm. Triplicate measurements were made for each sample and the averaged values were reported. The samples were incubated at 35 °C for 10 min followed by a gradual decrease in temperature to 2 °C at the rate of 0.5 °C per minute. T_{cloud} , the temperature for the onset of the phase separation as indicated by a rapid decrease in transmittance [15], was then determined. The liquid-liquid coexistence curve (or liquid-liquid phase diagram) was generated by plotting T_{cloud} against protein concentration.

2.5. Dynamic light scattering (DLS) for K_D determination

The mAbs were prepared in the desired solution and concentration then 0.2 μ m filtered. Each sample was spun at 2000 rpm for 2 min to remove any potential air bubbles before analysis. The samples were analyzed using a high-throughput 384-well plate DynaPro DLS instrument (Wyatt Technology, Santa Barbara, CA, USA) equipped with a 633 nm laser. The scattered light was monitored at 173° to the incident beam and auto correlation functions were generated. The mutual diffusion coefficient (D_m) obtained in DLS was plotted against the protein concentration (g/mL) to obtain the interaction parameter (K_D) from the following equation: $D_m = D_s (1 + K_D \times C)$ [23]. D_s is the self-diffusion coefficient (the value of D_m at infinite dilution), and C is the protein concentration. D_s is obtained from the intercept of a plot of D_m versus C , and K_D was obtained from the slope.

2.6. Viscosity measurement

The viscosities of the samples were measured using *m*-VROC™ viscometer (RheoSense Inc., San Ramon, CA, USA). Samples were loaded into a 0.5 mL syringe and allowed to reach thermal equilibrium at ambient temperature (22 °C) before measurement. Viscosities under four different shear rates (ranging from 100 to 1600 s⁻¹) were evaluated. The reported viscosity is average value from different shear rates and triplicate measurements.

2.7. Analytical size-exclusion chromatography (HP-SEC)

Purity analysis was performed using a TSK-GEL G3000SWXL column (7.8 mm $\times$ 30 cm) from Tosoh Bioscience (King of Prussia, PA, USA) with an Agilent 1200 HPLC system (Agilent Technologies, Santa Clara, CA, USA). The column was equilibrated at a flow rate of

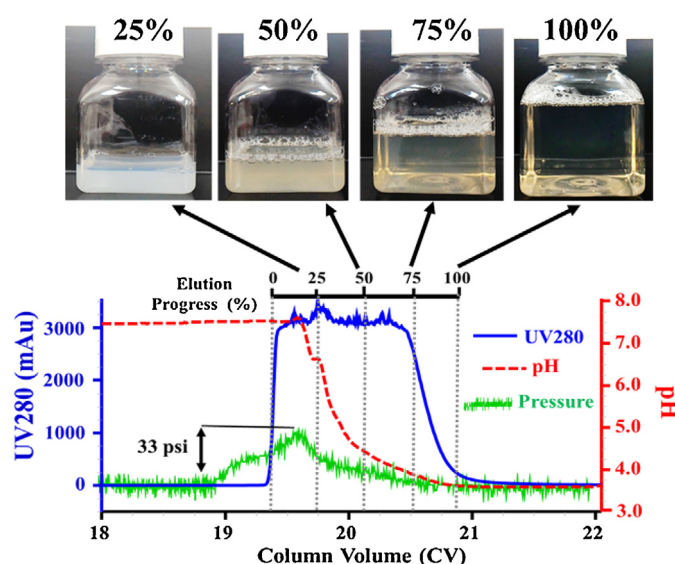


Fig. 1. Transient turbidity and pressure during the low pH elution process in Protein A chromatography. The UV₂₈₀, pH and pressure profiles for a mAb that exhibited turbidity and pressure increase during Protein A elution. Solution appearance of the elution pool at different progress stages during the elution process. The equilibration and elution buffers are 50 mM Tris-HCl, pH 7.4 and 25 mM NaAc, pH 3.6, respectively. The Protein A (MabSelect SuRe) column was loaded at 30 mg/mL load challenge. The results were obtained from a bench-scale run on AKTA Explorer platform without in-line filter.

0.5 mL/min with 78 mM sodium phosphate, 0.5 M sodium chloride, pH 7.4. All collected fractions were injected and eluted isocratically with the same buffer. The eluted protein was monitored by UV absorbance at 280 nm. Samples were 0.22 μ m filtered and injected at a mass load of 0.25 mg. All injected volumes were less than 0.7% of the column volume. The purity was estimated by integrating the chromatograms.

3. Results and discussion

3.1. Transient turbidity and pressure during low pH elution processes in Protein A chromatography

We observed an interesting elution turbidity issue in Protein A chromatography on many mAbs. An example is shown in Fig. 1 for an IgG1 mAb from a bench-scale run. The turbidity of the elution pool during the elution process is transient, i.e. the elution pool is highly turbid at the beginning of the low pH elution process and gradually becomes optically clear as the elution progresses. In these cases, both the step yield and the purity of the final elution product were not found to be affected (data not shown).

However, this transient turbidity resulted in high back pressure in both large- and small-scale Protein A column (33 psi in Fig. 1 from a small scale run). In these cases, elution flowrates have to be reduced to lower back pressure in order to continue purification. In Protein A purification skid for large scale purification, a depth filter and a bioburden reduction filter (0.2 μ m filter) are usually used in-line with the Protein A column to remove particulates and reduce bioburden of the eluted product, tending to exacerbate the high back pressure. When lowering flowrate alone is not sufficient to mitigate high back pressure, the depth filter and/or bioburden reduction filter are typically bypassed. The unfiltered elution products were collected and held in a temporary tank or container before being filtered off-line. These led to complicated processing, process deviations and increased bioburden contamination risk. While these observations provided better empirical understanding, the mechanism on turbidity formation remained

unclear. In the next section, we describe a series of experiments to identify the root cause of this behavior and gain fundamental understanding.

3.2. The turbidity is associated with neutral pH and high protein concentration

To select suitable mAbs for more in-depth bench-scale studies, we screened 12 mAbs (spanning the IgG1, IgG2 and IgG4 isotypes) on a bench-scale MabSelect SuRe Protein A column. To minimize the effects of host cell protein or improperly folded product, purified mAbs (>95% monomer purity by SDS-PAGE and <100 ng HCP/mg mAb; data not shown) were tested. As summarized in Table 1, half of the mAbs tested exhibited transient turbidity during the low pH elution process. These mAbs include the IgG1, 2 and 4 isotypes, suggesting antibody isotype does not play a role in the phenomenon. Among them, mAbs A, B and C showed the highest turbidity; therefore, were chosen as model proteins in our studies.

To elucidate which particular solution conditions are associated with the turbidity for the three mAbs, these elution products were collected according to UV₂₈₀ signal into 0.25 CV fractions (Fig. 2A). As shown in Fig. 2A, mAbs A, B and C had similar elution profiles, besides the A₂₈₀ measurement, the UV₆₀₀ signal was also recorded to monitor turbidity changes as mAbs typically do not absorb at the 600 nm wavelength. The three mAbs all showed a significant UV₆₀₀ peak (Fraction 2 indicated by red arrow in Fig. 2A) at an early stage of the low pH elution step. This increase in UV₆₀₀ is consistent with the onset of turbidity observed in fraction 2 (indicated by the red arrow in Fig. 2B), demonstrating the utility of the UV₆₀₀ signal to monitor turbidity changes during the elution process. Interestingly, the fraction corresponding to fraction 2 was the most turbid for each of the three mAbs evaluated.

To determine what is different between fraction 2 and the other fractions, the mAb concentration and pH for each fraction were measured and compared to each other. As shown in Fig. 2C, the turbid fractions from the three mAbs evaluated had the following common features: (1) relatively high protein concentrations: 25.6, 48.4 and 31.2 mg/mL for mAb A, B and C, respectively (indicated by red arrow in Fig. 2C); and (2) close-to-neutral pH values: pH 7.4, pH 7.2, and pH 6.8 for mAb A, B and C, respectively. While the next fraction for each individual mAb may have had higher protein concentration than seen in fraction 2, the pH was more acidic (pH 4.5–5.1) and surprisingly showed little to no turbidity. Our results are contrary to the multiple published studies suggesting that low elution pH and high protein concentration results in protein precipitation and turbidity [8–10]. For the three mAbs studied here, the cause of the turbidity appears to be associated with neutral pH and high protein concentration.

3.3. The elution turbidity in Protein A chromatography is caused by LLPS induced liquid droplets

To better understand the root cause of the observed turbidity, the three turbid fractions from the different mAbs were analyzed by micro fluid imaging (MFI). As shown in Fig. 3A, there are many liquid droplets detected within the three turbid fractions. These liquid droplets are spherical with various sizes. No protein particulate matter or precipitates (typically observed as irregular fibrils or clusters) were seen in these samples. Therefore, the turbidity is likely caused by liquid droplets as opposed to the commonly seen particulates. As shown in Fig. 3B, allowing the samples sit quiescently for 3 h at room temperature led to two liquid phases (left panel). Both the upper and the lower phases contain mAb, with the lower phase being highly concentrated (~140 mg/mL) and the upper phase at a much lower protein concentration (2–10 mg/mL). The turbidity of these three fractions was found to be tempera-

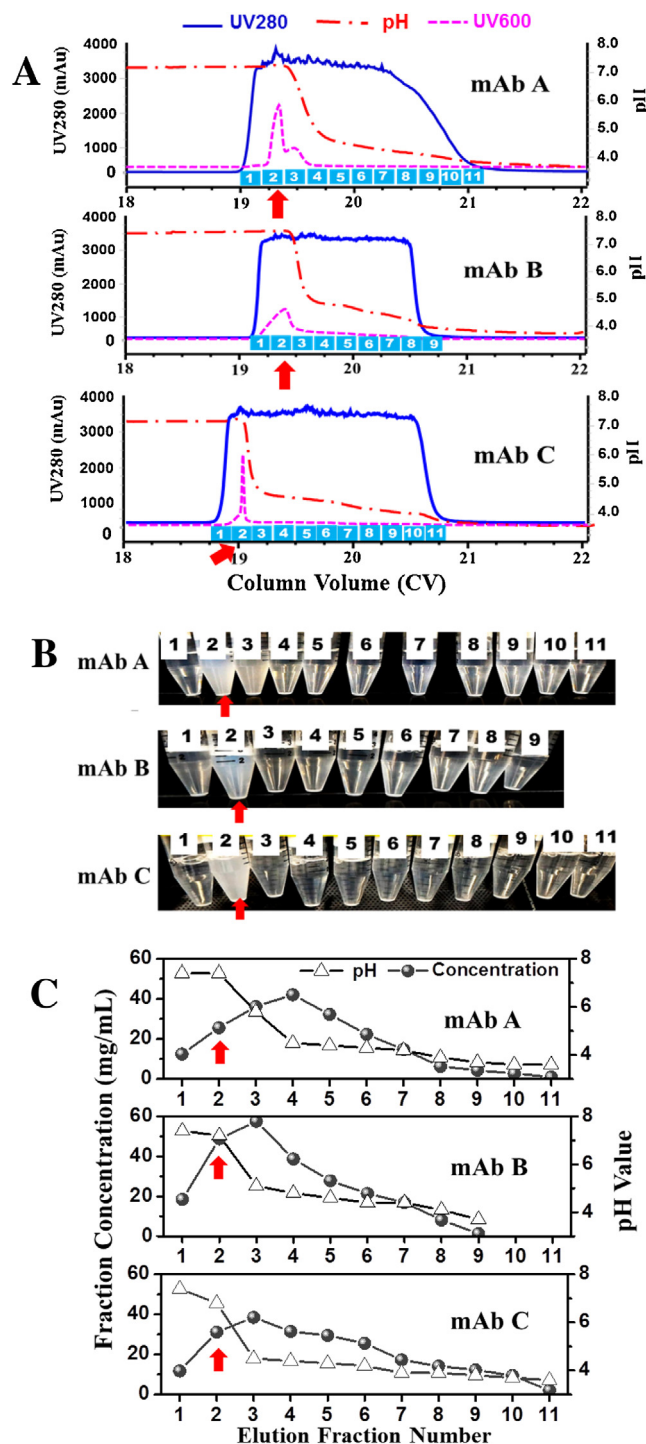


Fig. 2. Increased turbidity is associated with neutral pH and high protein concentration. (A) Protein A chromatograms for the three selected mAbs. (B) Solution appearance of the collected elution fractions. (C) Protein concentration and pH values for the collected fractions. The Protein A (MabSelect SuRe) column was loaded at 30 mg/mL load challenge.

ture dependent: when the samples were warmed up to 30 °C and mixed, they all became clear (Fig. 2B, middle panel), and the turbidity returned when these 30 °C samples were chilled to 4 °C (Fig. 3B, right panel).

Based on the observations of liquid droplets, temperature-dependent turbidity and two liquid phases, we surmise that liquid–liquid phase separation (LLPS) [14,16–21] occurred in the three samples. The turbidity of each sample was further studied

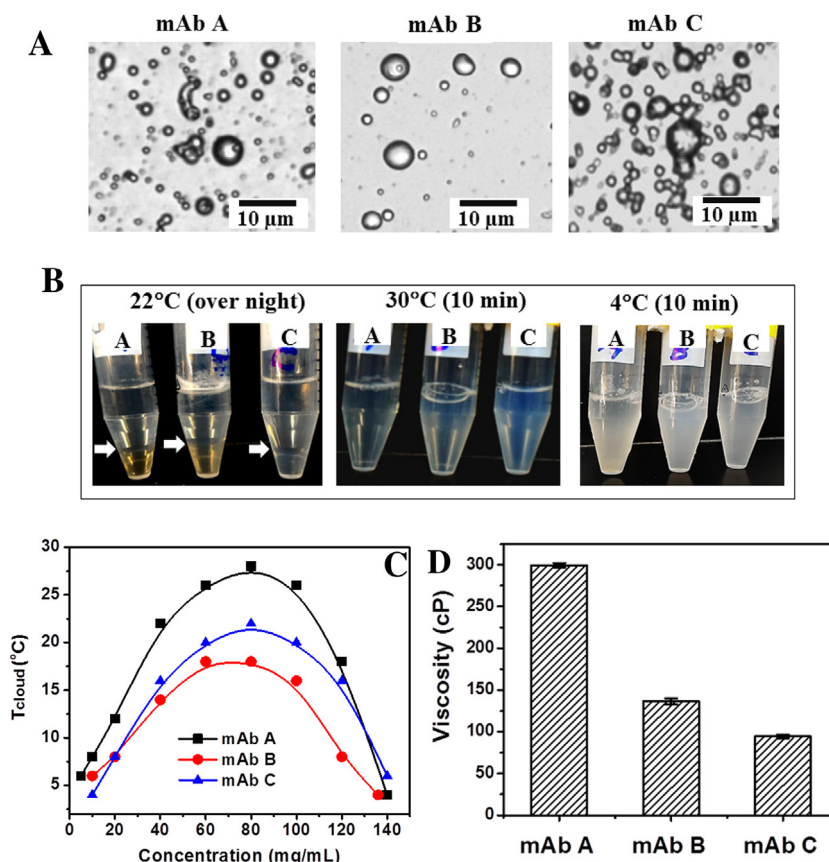


Fig. 3. The transient turbidity is caused by LLPS-induced dense phase liquid droplets. (A) MFI images for the three turbid samples. The sample turbidity is caused by dense phase liquid droplets in the solutions. (B) The appearance of the turbid solutions at 22, 30 and 4 °C. The three samples demonstrated temperature-dependent turbidity. The three samples in (A) and (B) are in the mixed of 50 mM TrisHCl and 25 mM NaAc buffers with pH value around 7.0. Protein concentrations of these samples are 23, 48, and 30 mg/mL, respectively. (C) Liquid-liquid coexistence curves for the three mAbs. (D) Viscosities for the dense phases for the three mAbs. Samples in (C) and (D) are in 50 mM TrisHCl pH7.4.

by monitoring UV_{600} transmittance at temperatures of 35–10 °C. The transmittance changed significantly at 26, 15 and 20 °C for mAb A, B and C, respectively (data not shown). These temperatures represent a key parameter for describing LLPS, namely T_{cloud} . The plotting of T_{cloud} against protein concentration constructs the liquid-liquid coexistence curve in a temperature-concentration phase diagram [15]. Conditions under the coexistence curve lead to LLPS as characterized by a turbid solution initially and then separation into two stable liquid phases; conditions above the curve lead to homogeneous, optically clear solutions. As discussed above, the instantaneous concentration of the elution fraction can exceed 30 mg/mL concentration in Protein A chromatography. The results in Fig. 3C explain that, for mAbs A, B, and C, LLPS induced turbidity would be expected to occur during the elution process in the typical manufacturing temperature range (15–25 °C).

Since the turbidity for the three mAbs is not caused by precipitates but, rather, liquid droplets, why did the column pressure increase during the Protein A elution of the three mAbs? We hypothesized that it may be due to high viscosity of the high protein concentration containing liquid droplets (140 mg/mL). To test our hypothesis, the viscosity readings of the dense phase (settled droplets) from the three mAbs were measured. As shown in Fig. 3D, the dense phases (at ~140 mg/mL) showed extremely high viscosity: 280, 120 and 80 cP for mAb A, B, and C, respectively. As what we discussed above, the dense phase is formed by the settled down liquid droplets. These liquid droplets therefore are expected to have the same viscosity as the dense phase. Based on the results, we conclude that the high viscosity of the liquid droplets and the narrow

channels inside the Protein A beads contribute to the significant column back pressure.

In addition to these three model mAbs, as summarized in Table 1, several other tested mAbs also exhibited LLPS in Protein A chromatographic screening. Furthermore, Chen et al. recently reported that an IgG2a (under neutral pH condition) liquid-liquid phase separated upon refrigerated storage, indicating that this phenomenon may be more wide-spread than previously believed [23]. Furthermore, Trilisky et al. conducted a systematically evaluation on 20 mAbs and found that all of those mAbs exhibited LLPS under certain conditions [24]. They found that some of these conditions are similar to what is expected during chromatography, particularly in bind-elute mode, where local protein concentrations in the column can be very high [24]. Taken together with our results, it is likely LLPS occurs frequently during Protein A chromatography in mAb downstream processing. The under-reporting in the literature is possibly due to the transient and reversible nature of LLPS.

3.4. Characterization of the LLPS behaviors for the 3 mAbs

There were no significant differences observed between the proteins detected in the two phases as determined by circular dichroism, capillary isoelectric focusing (cIEF), size exclusion chromatography, and bioassay for activity (data not shown). These results are consistent with what Rout et al. who reported on a dual variable domain IgG [19–21]. The same study also suggested that the LLPS of the mAb is an equilibrium process that is reversible. In our work, to test whether the LLPS behavior of the three mAbs is reversible in nature, the three mAbs were prepared at different con-

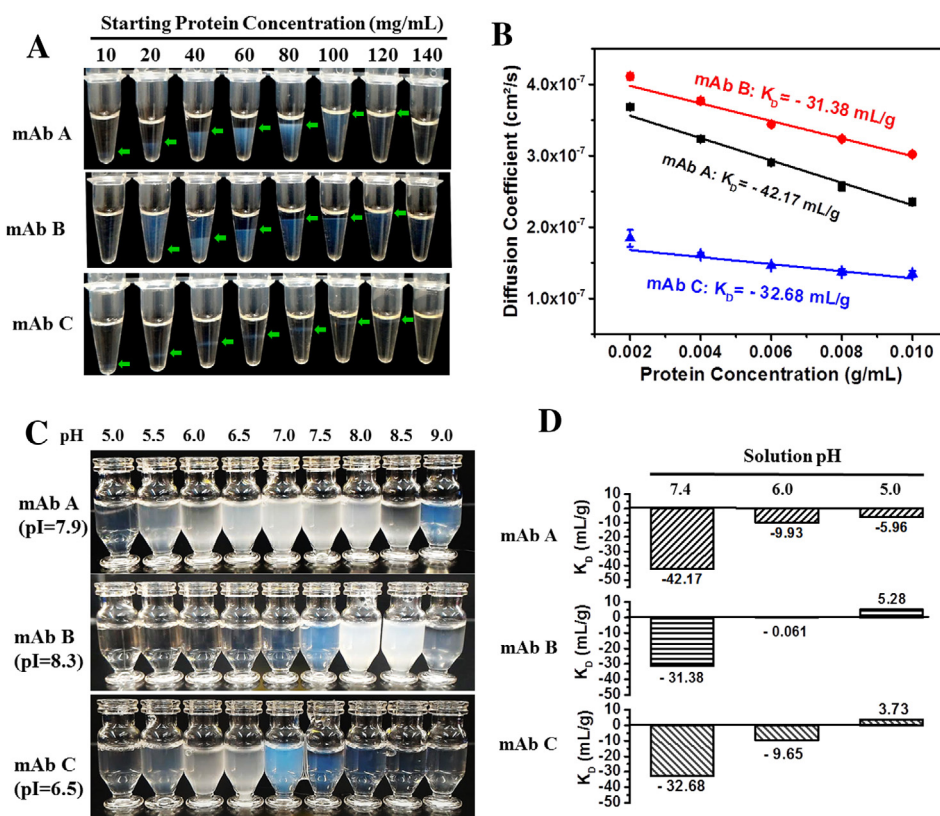


Fig. 4. Characterization of the LLPS behaviors for the three mAbs. (A) The LLPS of the three samples is reversible equilibrium process. The samples from the dense phase were diluted in buffer A in a series of protein concentrations, and the diluted sample was allowed to reach equilibrium at 4 °C. (B) Plotting of diffusion coefficients against protein concentrations the three mAbs to obtain the interaction parameters (K_D s). The samples were prepared at the desired concentration in 50 mM Tris-HCl pH7.4. (C) Solution appearance for the three mAbs prepared at different pH values. Protein concentration for each sample is around 30 mg/mL. The initial samples were prepared in 5 mM NaAc pH 5.0 and then 1 M Tris base pH 11.0 was used to get the desired pH value. Tris-HCl concentration in these samples ranged from 5 to 70 mM. The turbid samples all phase separated into two liquid phases after 3 h. (D) The interaction parameters (K_D s) for the three mAb in the following solution: 50 mM Tris-HCl pH 7.4; 50 mM His-HCl, pH 6.0; 50 mM NaAc pH 5.0. DLS measurement was conducted at 25 °C.

centrations in 50 mM Tris-HCl buffer. After phase separation was completed, the liquid–liquid interface (indicated by green arrows) clearly indicates that the samples with higher initial concentrations have a larger dense phase. These results demonstrate the LLPS for mAb A, B, and C is a reversible equilibrium process, consistent with the previously published results [14,16–21].

Recently, several publications have suggested LLPS correlates with protein self-association [14,18–21,25]. To test whether it is the case for the three mAbs in this study, the diffusion coefficients at different protein concentrations were obtained by dynamic light scattering (DLS). The diffusion coefficients for different protein concentrations were plotted and shown in Fig. 4B. The diffusion coefficient decreased with increasing protein concentration, suggesting self-association. The interaction parameters (K_D) were derived by plotting diffusion coefficient against protein concentration. The K_D values at pH 7.4 for mAb A, B, and C are -42.17 , -31.38 , -32.68 mL/g , respectively (Fig. 4D). The results suggest that mAb A, B and C are strongly self-associating under those conditions, leading to the observed phase separation behavior of these mAbs, consistent with the published results [14,18–21,25].

Similar to protein concentration, we also hypothesized that solution pH may also contribute to LLPS as solution pH determines the total charge of the mAbs, potentially impacting self-association. To test our hypothesis, each mAb was prepared in a different pH solution. As shown in Fig. 4C, the sample from each mAb was most turbid when the solution pH is close to the protein pI; when solution pH is far away (higher or lower) from protein pI, the solution becomes less turbid or optically clear. All turbid samples separated

into two liquid phases after 3 h quiescent hold, suggesting the turbidity was driven by LLPS. Therefore, solution pH has significant impact on LLPS. The solution pH determines protein charge status: when the sample pH is close to protein pI, the molecule as a whole carries no net charge, i.e. minimal electrostatic repulsion; when pH is below the pI, the protein is positively charged, i.e. electrostatic repulsion. It appears electrostatic interactions play a key role in the self-association of the three mAbs under study. To test this hypothesis, K_D value of the three mAbs at different pH values were determined by DLS (Fig. 4D). The self-association becomes weaker at pH 6.0, compared to pH 7.4; at pH 5.0, the K_D for mAb A approaches 0 mL/g, and the K_D for mAbs B and C become positive (i.e. repulsive interactions). Based on these results, it can be concluded that the LLPS of the three mAb is concentration-dependent equilibrium process that is due to protein self-association. Solution pH has significant impact as electrostatic interactions dominate self-association under conditions evaluated for the three mAbs.

3.5. The effects of NaCl and Arg-HCl on LLPS-induced turbidity

The ionic strength of the protein solution is also expected to significantly impact LLPS behavior, and optimizing solution ionic strength should enable mitigation of the LLPS-induced turbidity. To test this hypothesis, each of the three mAbs was prepared in solutions with different concentrations of NaCl. As shown in Fig. 5A, the presence of NaCl effectively reduces sample turbidity, where 50 mM NaCl was sufficient to make all the three mAb samples change from turbid to optically opalescent. At 100 mM NaCl, mAb B

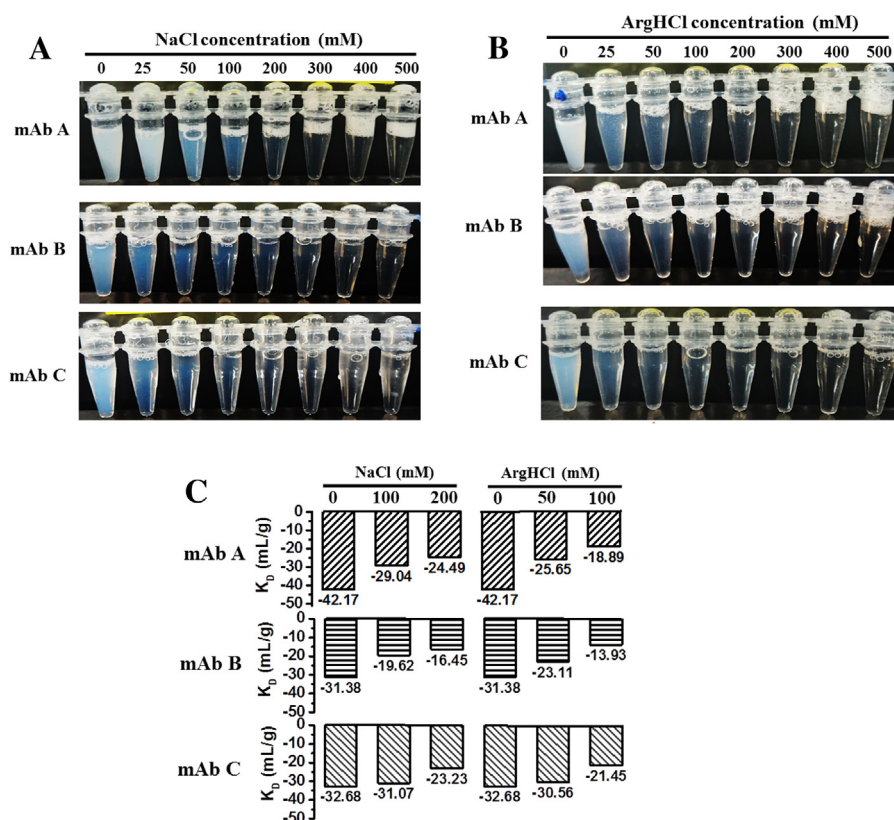


Fig. 5. The addition of NaCl or Arg-HCl mitigates LLPS and weakens self-association. The effects of different concentrations of NaCl (A) or Arg-HCl (B). Each sample contains 30 mg/mL protein. The buffer was 50 mM Tris-HCl, pH 7.4 and the desired amount of NaCl or Arg-HCl was prepared in 50 mM Tris-HCl, pH 7.4. (C) The effect of NaCl or Arg-HCl on the association parameters (K_D s).

and C become optically clear, while mAb A required 200 mM NaCl to become optically clear.

Arginine-HCl (Arg-HCl) has also been reported in the literature to be able to mitigate protein self-association [26,27]. Therefore, Arg-HCl was also tested. As shown in Fig. 5B, like NaCl, Arg-HCl can also effectively reduce sample turbidity. The presence of 100 mM Arg-HCl was sufficient to make the three mAbs solutions become optically clear.

As shown in Fig. 5C, the K_D values from the corresponding conditions revealed that the presence of NaCl or Arg-HCl results in less negative K_D , i.e. self-association is weakened by the presence of NaCl or Arg-HCl. Arg-HCl is more effective than NaCl on reducing turbidity and self-association. Arg-HCl not only increases ionic strength but also has the guanidinium side chain which preferentially binds to hydrophobic residues to inhibit self-association [26,27]. Therefore, besides the dominant electrostatic interactions, hydrophobic interaction may also contribute to the LLPS of the three mAbs. More importantly, addition of NaCl or Arg-HCl is an effective method for mitigating LLPS-induced turbidity.

3.6. The effects of elution buffer composition on the elution turbidity

As discussed above, the presence of NaCl or Arg-HCl effectively mitigated LLPS-induced turbidity in bulk solution. To test whether they reduce elution turbidity in Protein A chromatography, elution solutions containing different concentrations of NaCl or Arg-HCl were tested. As shown in Fig. 6A and B, the addition of NaCl or Arg-HCl to the elution buffer effectively decreased the UV₆₀₀ peak, representing a decreased turbidity during the elution process. The elevated UV₆₀₀ peak was fully mitigated by addition of 100 mM NaCl or 50 mM Arg-HCl. These results are consistent with the

previous bulk solution study results that the addition of NaCl or Arg-HCl reduced the LLPS-induced turbidity. As discussed above in the Sections 3.4 and 3.5, the LLPS-induced turbidity is driven by protein self-association and the addition of NaCl or Arg-HCl to the solution results in weaker self-associations. Additionally, the addition of NaCl or Arg-HCl showed subtle effects on pH transition that may contribute slightly to the turbidity changes. The product purity as measured by size-exclusion chromatography was unchanged by addition of NaCl or ArgHCl (data not shown). However, one potential drawback in adding NaCl or Arg-HCl to the elution buffer is that the elution product had higher conductivity (~10 mS/cm for 100 mM NaCl and ~5 mS/cm for 50 mM Arg-HCl), which could impact downstream operations and process integration. For instance, removal of process impurities is impaired on a flow-through anion exchange chromatography (AEX) as NaCl or Arg-HCl modulates electrostatic interactions; similarly the binding capacity of cation exchange chromatography (CEX) is decreased. To eliminate these impacts, extra conditioning, such as dilution of the column load with water, may be needed. However, dilution steps (especially high dilution ratios) are not favorable for large-scale manufacturing as it can present facility fit issues and increase processing time. During the low pH elution process in Protein A chromatography, the early part of the elution pool is at a neutral pH because the product begins to elute prior to the pH wave exiting the column. The temporal difference between product elution and the pH wave depends on the compositions of the preceding wash and elution buffers. As described above, the LLPS-induced turbidity was reduced by decreasing the solution pH from neutral to acidic (Fig. 4C). An earlier transition from neutral equilibration to the acidic elution solution should be able to mitigate the transient turbidity increase observed during Protein A chromatography, as would be the case when using an elution buffer with higher buffer

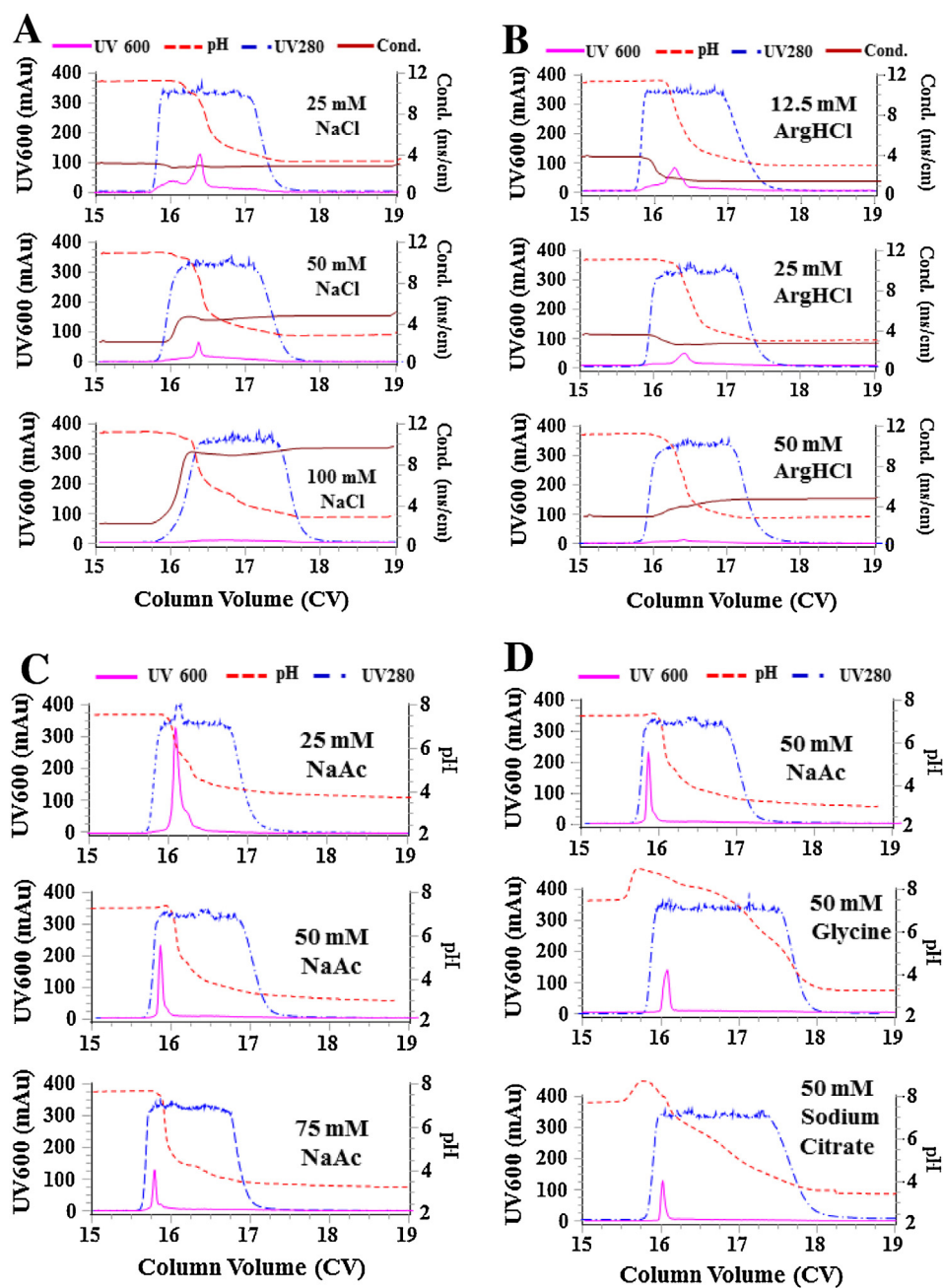


Fig. 6. The effect of elution buffer composition on the Protein A elution turbidity (A) Different concentrations of NaCl in 25 mM NaAc, pH3.6 as the elution buffer. (B) Different concentrations of Arg-HCl in 25 mM NaAc, pH3.6 as the elution buffer. (C) Different concentrations of NaAc at pH 3.6 as the elution buffer. (D) Protein A chromatography elution by 50 mM NaAc, glycine, or sodium citrate (all at pH3.6). MabSelect SuRe resins and mAb A were used in the experiments. All experiments used 30 mg/mL resins load challenge.

concentration. We tested sodium acetate (NaAc) elution buffer at 25, 50 and 75 mM for the Protein A elution buffer. Higher concentration elution buffer led to earlier pH wave transition (Fig. 6C). The UV₆₀₀ peak was smaller and the column back pressure was lower as the buffer concentration increased, again demonstrating a reduction in transient turbidity in the effluent stream. Besides the earlier pH wave transition, the higher ionic strength should also help as discussed previously. The elution turbidity still persisted at 75 mM NaAc, which provided an early pH wave transition. We surmise NaAc concentration higher than what we evaluated in this study would further reduce the elution turbidity; however, using higher NaAc concentrations is not ideal, as significant dilution may occur

in the downstream unit operations such as pH adjustment for low pH viral inactivation and pH conditioning for CEX.

In addition to NaAc, we also tested two other elution buffer types that can be used as elution solutions for Protein A chromatography, glycine and sodium citrate. The three buffers at the same concentration (50 mM) and pH value (pH3.6) were tested on a Protein A column and the corresponding chromatograms were shown in Fig. 6D. As can be seen in Fig. 6D, NaAc, glycine, and citrate buffers generated pH transitions that were quite different in shape and duration. Compared to NaAc, glycine and sodium citrate showed significant larger pH transition phase (in volume) going from pH 7.4 to pH 3.6, with corresponding significant larger elution pool volume. Therefore, for glycine and sodium citrate, the elution

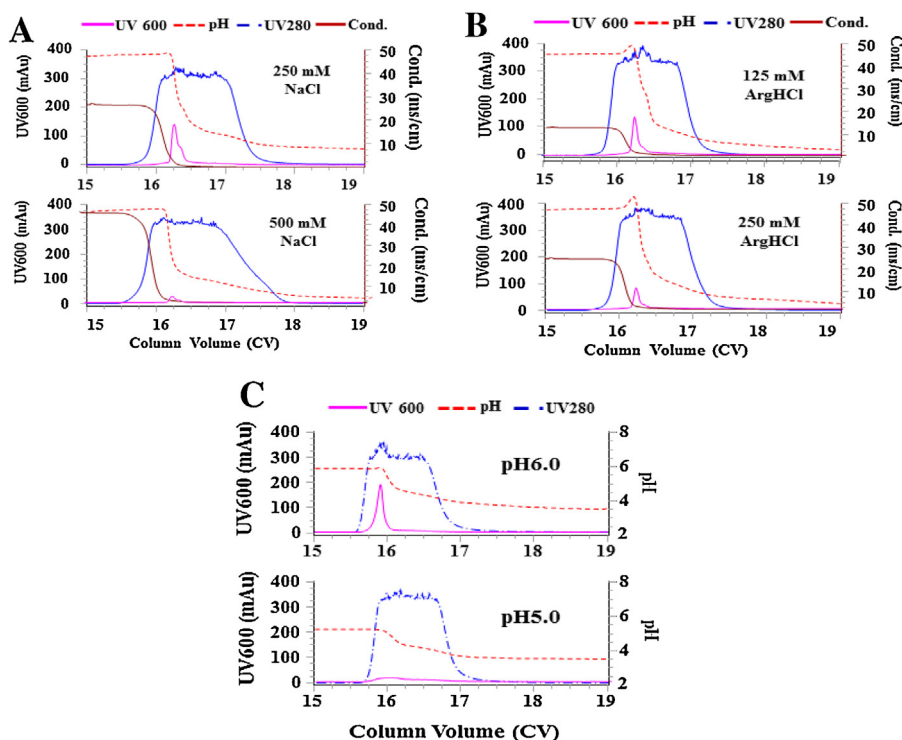


Fig. 7. The effects of re-equilibration buffer composition on elution turbidity in Protein A chromatography. (A) Different concentrations of NaCl in 50 mM Tris-HCl, pH 7.4 as re-equilibration buffer. (B) Different concentrations of Arg-HCl in 50 mM Tris-HCl, pH 7.4 as re-equilibration buffer. (C) 50 mM His-HCl, pH 6.0 and 50 mM NaAc pH 5.0 as the re-equilibration buffer. MabSelect SuRe resins and mAb A were used in the experiments. All experiments used 30 mg/mL resins load challenge.

protein concentration is lower but the exposure to neutral pH is longer. As discussed above, lower protein concentration is unfavorable for LLPS but neutral pH promotes LLPS. These two factors determine the LLPS-induced elution turbidity. As showed in Fig. 6D, glycine and sodium citrate led to slightly smaller UV₆₀₀ peak (elution turbidity), compared to NaAc. These results suggest the effects of shallow pH transition are offset by those of lower protein concentration. It is noteworthy that the three buffers at same 50 mM concentration have different ionic strengths which may contribute to pH transition and LLPS-induced turbidity. More thorough comparison can be conducted to scrutinize the small difference among the three buffers. However, based on our results, it can be concluded the three buffers all lead to elution turbidity issue and a different buffer species does not seem to solve the issue.

3.7. The effects of re-equilibration buffer composition prior to elution on turbidity

As the turbidity occurs during the transition from the re-equilibration buffer to the elution buffer, the re-equilibration buffer composition is another potential means to mitigate elution turbidity. We tested the effects of different concentration of NaCl or Arg-HCl added to the re-equilibration buffer (50 mM Tris-HCl, pH 7.4) on a Protein A process using mAb A. As shown in Fig. 7A and B, 500 mM NaCl or 250 mM Arg-HCl in the re-equilibration buffer resulted in a smaller turbidity peak and lower back pressure. Therefore, addition of NaCl or Arg-HCl to the re-equilibration buffer is also effective solution for mitigating elution turbidity. However, the resulting conductivity of the elution product was found to be higher for both NaCl and Arg-HCl due partial incorporation of the re-equilibration buffer in the elution pool. Additionally, when a high concentration of NaCl or Arg-HCl was used, complete elution required a larger elution volume. This may be explained by the findings from Bolton et al. that the presence of high salt con-

centration promotes hydrophobic interactions between Protein A and the mAb, requiring more acidic pH for mAb elution [28]. When high concentration of NaCl or Arg-HCl was used, a slightly slower pH transition was observed, which may also contribute to the larger elution volume. The reason for the slower transition is beyond the scope of our study and the titration of the resin ligands may be a factor. Another means to mitigate elution turbidity by modifying the preceding re-equilibration buffer is using a lower pH value for re-equilibration buffer prior to the low pH elution step since the acidic pH can prevent LLPS from occurring. To test this feasibility, different pH re-equilibration buffers were tested. As shown in Fig. 7C, lower pH buffer could effectively mitigate the UV₆₀₀ turbidity peak, especially when 50 mM NaAc, pH 5.0 was used, where the UV₆₀₀ turbidity peak was negligible. Mildly acidic pH buffer is often used as wash buffer to remove host cell proteins [6] and typically does not trigger protein elution. The conductivity of the product is low (<2 mS/cm) when this wash strategy is employed. The elution product yield and purity are comparable to the baseline run (data not shown). Taken together, a re-equilibration buffer at mildly acidic pH (such as pH 5.0) is a suitable and effective solution for mitigating transient elution turbidity.

3.8. The effects of load challenge, temperature, flow rate and Protein A resin type on the transient elution turbidity

Besides the solution conditions used to equilibrate and elute the mAb from the Protein A resin, several other factors, such as load challenge, operating temperature, flow rate for elution process, and type of Protein A resin were evaluated for impact on transient turbidity and LLPS. Lower load challenge resulted in lower transient turbidity (smaller UV₆₀₀ peak) during the low pH elution (Fig. 8A). This is expected because LLPS is protein concentration dependent. However, lowering load challenge is not an optimal solution for downstream processing because the lower throughput generally

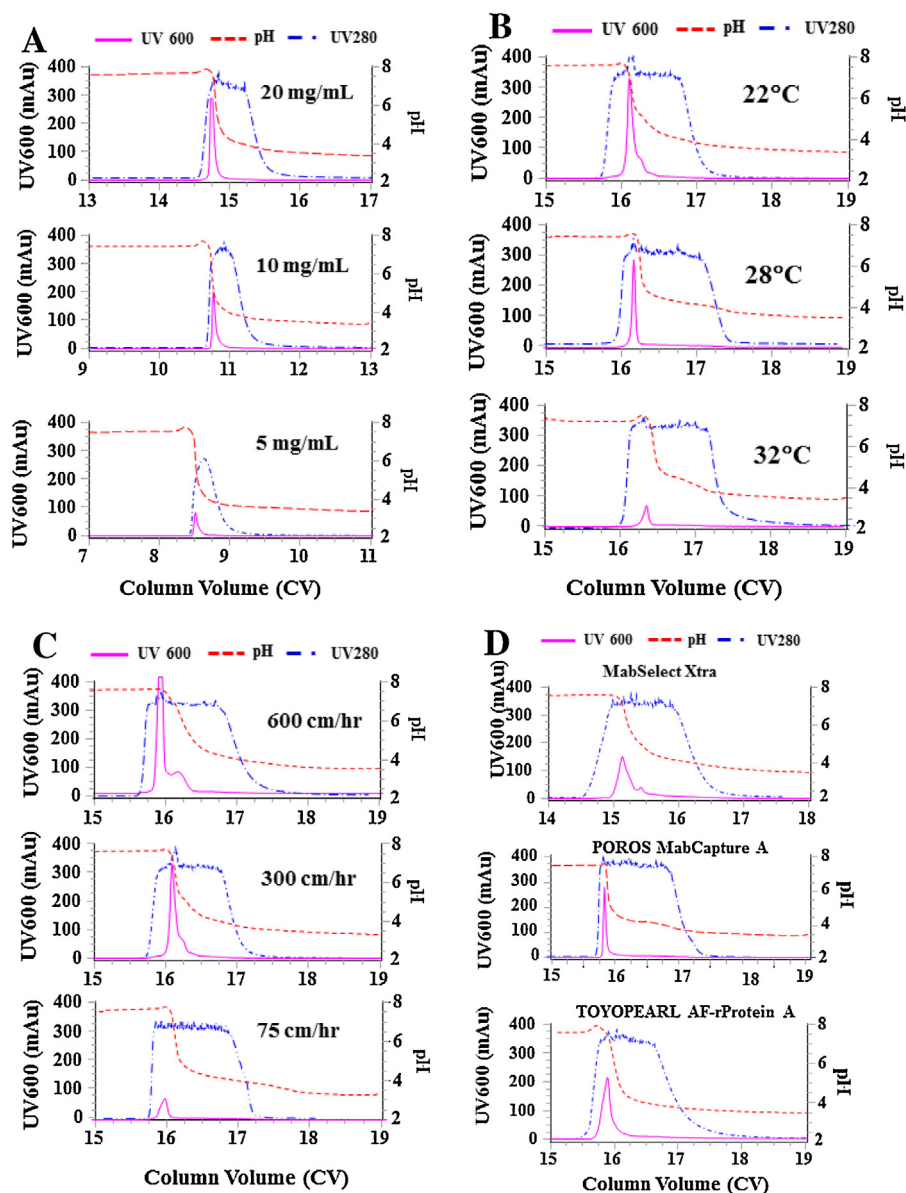


Fig. 8. The effects of Protein A process parameters on elution product turbidity due to (A) load challenge, (B) temperature, (C) flow rate, and (D) Protein A resins.

leads to a higher cost of goods for manufacturing (COGM) with larger size columns and/or more purification cycles.

Operating temperature was tested next, as it is expected to have minimal impact on process throughput or COGM. As expected, higher temperature led to lower turbidity (Fig. 8B), which is consistent with the trend discussed above that LLPS can be mitigated by raising temperatures. Furthermore, higher temperature may also help reduce solution viscosity in the absence of LLPS [23]. Operating a Protein A column at higher temperature appears to be a viable solution. However, at process scale, operating chromatography steps at temperatures higher than the typical manufacturing suite temperature (15–25 °C) is unattractive due to the need for specialized equipment such as jacketed columns and heat exchangers. In addition, the potential for higher degradation rates at elevated temperature is a concern.

The flow rate for the low pH elution process was also evaluated. As shown in Fig. 8C, lower flow rate resulted in lower elution turbidity. Lower flow rate resulted in sharper pH transition (red dotted line in Fig. 8C) as measured by elution volume and minimized the protein's exposure to neutral pH. As discussed above, LLPS is

mitigated at acidic pH. Therefore, the sharper pH wave transition occurring due to the lower flow rate decreased the elution turbidity. It would lead to longer processing time and smaller elution product volume. Since only the flow rate for elution needs to be lowered and smaller product volume usually does not have negative impact on manufacturing, lower flow rate is a feasible mitigation approach. Additionally, lower flow rate is also an easy mitigation method to implement in process-scale purification.

Three other Protein A resins (MabSelect Xtra, POROS MabCapture A, and Toyopearl AF-rProtein A) were also tested at the same load challenge to evaluate the role of different Protein A resins on the transient turbidity phenomenon. Similar to MabSelect SuRe, these three Protein A resins also exhibited elution turbidity (Fig. 8D) although their UV₆₀₀ profiles are different. We believe these differences are attributable to properties such as the structure and density of the Protein A ligand, mass transfer rate, bead size, and other factors specific to the individual resins that would tend to impact pH wave transition and instantaneous protein concentration during elution.

4. Conclusion

Turbid elution pools and high pressure are a common problem in Protein A chromatography during low pH elution. Historically, this behavior has been attributed to low pH instability and precipitation. In this work, we propose a new mechanism that may explain some of these observations. Through a systematic evaluation of three different mAbs, we determined that this transient turbidity is most likely caused by LLPS. The dense phase liquid droplets consist of high concentration mAb proteins and are also highly viscous, resulting in high column back pressure during the elution process.

Lower load challenge, higher operating temperature, lower elution flowrate, higher sodium chloride concentration in the elution buffer, addition of arginine in the elution buffer, increasing elution buffer concentration, or use of a mildly acidic wash buffer preceding elution are all viable options demonstrated in this work to control turbidity and column back pressure. Our results revealed that the use of a mildly acidic buffer preceding elution is the most suitable solution for reducing the elution turbidity in large-scale.

Our results also suggest that it is likely LLPS occurs frequently during Protein A chromatography in mAb downstream processing. The under-reporting in the literature is possibly due to the transient and reversible nature of LLPS. However, consistent with widespread occurrence, the signatures of LLPS are much more commonly observed, such as transient high column back pressure and effluent turbidity. Furthermore, we found higher load challenge in Protein A chromatography caused higher effluent turbidity as LLPS is protein concentration dependent. Continuing attempts to maximize the binding capacity of Protein A chromatography resins and binding capacities higher than 60 mg/mL have been reported [28]. High capacity protein A resins likely exacerbate the LLPS-induced elution effluent turbidity issue, suggesting a challenge for the future high capacity Protein A chromatography.

In our study, examples of IgG1, IgG2 and IgG4 mAbs all exhibited LLPS behavior. Thus, LLPS induced transient turbidity is unlikely to be associated with a specific IgG subclass. As expected, for the mAbs we evaluated LLPS was correlated with self-association and protein pI (Table 1). A self-associating mAb with a neutral pI is more likely to encounter transient turbidity and high column back pressure during Protein A chromatography.

Our study sheds new light and provides mechanistic understanding linking protein physiochemical properties and liquid–liquid phase separation behavior with process performance during Protein A chromatography. With this understanding, it should be possible to predict whether a given mAb will demonstrate LLPS during Protein A elution as well as develop a proactive mitigation strategy.

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